



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-03 Start Here

Orchestration & Reliable Workflows - learn Airflow as an operator-safe workflow coordinator, not as a magic scheduler.

Artifact	C3-03
Status	2026 - LEARNER EDITION
Phase	C3-03 Orchestration & Reliable Workflows 13 lessons 34-46 focused hours
Prerequisite	C3-02 Gate PASS; C2B Python/quality/ownership foundations assumed, orchestration taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: simple mental picture -> tiny example -> guided Airflow/tool workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A green scheduler box alone never proves correct data.



What this phase changes

C2B and C3-00/01/02 taught reusable Python, validation, ingestion and transformation. C3-03 teaches how to coordinate those durable units across time, failure, waiting, retries, historical reprocessing and constrained resources.

This is not a duplicate of Bridge. Orchestration is a new engineering layer; every lesson is deeper engineering progression.

Exact learner loop

- Study the 3-page C3-040 lesson.
- Reproduce the guided example with lesson open.
- Close lesson; do matching Pxx in C3-041.
- Save code/config/output + confidence + main uncertainty + independent validation.
- Only then open matching Rxx in C3-042.
- Close review and complete the changed retry.
- Mark Mastered only when the fresh changed task passes and evidence is saved.

Phase sequence

ID	Lesson
C3-03-L01	Workflow orchestration and DAG design
C3-03-L02	Airflow setup, standalone mode and project structure
C3-03-L03	DAG authoring with TaskFlow and tasks
C3-03-L04	Schedules, logical dates and data intervals
C3-03-L05	Parameters, variables, connections and secret boundaries
C3-03-L06	Retries, timeouts and failure callbacks
C3-03-L07	Catchup, backfills and controlled reprocessing
C3-03-L08	Sensors and data-aware dependencies
C3-03-L09	Idempotent task design and partial reruns
C3-03-L10	Pools, concurrency and resource limits
C3-03-L11	DAG testing and pre-deployment validation
C3-03-L12	Operational logs, metadata and troubleshooting
C3-03-L13	Deployment patterns, recovery runbooks and handoff



Assessment and runtime honesty

After all 13 lesson mastery records, complete RapidRoute. Then attempt MetroParcel Gate A. If a narrow competency fails, use only the assigned remediation card and a changed micro-retest. Gate B/C remain sealed unless explicitly assigned as a fresh full retest.

PASS means every required competency passes and there is zero unresolved critical failure. A failed required competency cannot be offset by stronger performance elsewhere.

Live Airflow boundary

This artifact build can execute pure Python pipeline logic, pytest and static AST checks. It does not have Airflow installed, so scheduler/UI/DagRun, sensor wait/timeout, pool queueing and actual backfill commands must be proven on the learner WSL2 environment. Those are deliberately NOT CLAIMED here.

Mentor route contract

Common shell remains: Today | Learn | Course | Repair | Evidence | Jobs | Reports | Settings. Each lesson route will point to an exact learner-book page, practice page, review page and relevant starter ZIP/file.